

Problem 9

Problem 9

Problem 1 A piecewise defined function f is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Since $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2}$, the form of the limit is $\frac{\#}{0}$.

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{2x-3}{x-2} = -\infty,$$

since the numerator positive, the denominator negative and goes to 0. Therefore, the line $x = 2$ is a vertical asymptote of f since $\lim_{x \rightarrow 2^-} f(x) = -\infty$.

Note: The line $x = -2$ is not a vertical asymptote, since for values of x near -2 , $f(x) = \frac{2x-3}{x-2}$. So, f is continuous at -2 .

- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. Check End Behavior as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} \frac{x^2-5x+6}{x^2-4} \text{ is of the form: } \frac{\infty}{\infty}$$

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$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{x^2 - 5x + 6}{x^2 - 4} &= \lim_{x \rightarrow \infty} \frac{x^2 - 5x + 6}{x^2 - 4} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{1 - \frac{5}{x} + \frac{6}{x^2}}{1 - \frac{4}{x^2}} \\ &= 1\end{aligned}$$

Therefore, the line $y = 1$ is a horizontal asymptote of f since $\lim_{x \rightarrow \infty} f(x) = 1$.

Check End Behavior as $x \rightarrow -\infty$:

$\lim_{x \rightarrow -\infty} \frac{2x - 3}{x - 2}$ is of the form: $\frac{\infty}{\infty}$

$$\begin{aligned}\lim_{x \rightarrow -\infty} \frac{2x - 3}{x - 2} &= \lim_{x \rightarrow -\infty} \frac{2x - 3}{x - 2} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} \\ &= \lim_{x \rightarrow -\infty} \frac{2 - \frac{3}{x}}{1 - \frac{2}{x}} \\ &= 2\end{aligned}$$

Therefore, the line $y = 2$ is a horizontal asymptote of f since $\lim_{x \rightarrow -\infty} f(x) = 2$.